

**NYU**SCHOOL OF GLOBAL
PUBLIC HEALTH

Intermediate Epidemiology

Lecture 1 - Introduction

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Lecture objective

1. To discuss course objectives, structure and assignments
2. To REVIEW measures used in describing health status in a population

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Course Overview

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Overall course objectives

- Over the course of this semester, we will:
 - Build on foundational concepts of epidemiology
 - Conduct a descriptive analyses to investigate an epidemiologic question
 - Learn how to communicate – explain and interpret epi results in both oral and written formats!

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Course Structure

- Two main components:
 - Lecture (weekly, Tuesdays, in-person)
 - Recitation (lab) sessions (weekly) – required!
- Evaluation components:
 - Individual research project
 - Recitation section exercises and attendance
 - Include Stata exercises
 - Recitation materials will be available on the main course site
 - Final exam

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Analysis Project

GPH GU 2450 - Intermediate Epidemiology Analysis Project – SPRING 2026

OVERVIEW

During this semester students will plan, analyze, interpret and prepare reports of findings from an epidemiologic research investigation. Students will receive feedback on these assignments so that they are prepared to conduct the analyses and prepare a report. All components of this exercise are detailed in the instructions presented in this document.

DATA

For this assignment, the instructor will provide a datasets from the August 2021 – August 2023 National Health and Nutrition Examination Survey (NHANES), administered by the Centers for Disease Control and Prevention (CDC). In addition, you will be provided with a codebook that provides information on all relevant variables.

TOPIC

This semester, the main outcome of interest will be **depression**. Students should review the following *Morbidity and Mortality Weekly Report*, published by the CDC, to gain additional background and understanding of recent epidemiological patterns in depression among adults in the U.S.:

Lee B, Wang Y, Carlson SA, et al. National, State-Level, and County-Level Prevalence Estimates of Adults Aged ≥18 Years Self-Reporting a Lifetime Diagnosis of Depression — United States, 2020. *Morbidity and Mortality Weekly Report* 2023 June 16;72(24):644-650.

Assignment

Analysis project

- A. Research question & hypothesis
- B. Annotated Bibliography and Epidemiological trends
- C. Methods section
- D. Univariable analysis + write up
- E. Bivariable analyses + write up
- F. Poster presentation – preparation
Poster presentation – oral presentation
- G. Final write-up + multivariable analysis (abstract, introduction, methods, results, discussion)

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Measures in epidemiology

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Overview

- This lecture will describe the following measures and methods:
 - Open and closed cohorts
 - Measures of prevalence
 - Counts, proportions, odds
 - Measures of incidence
 - Survival probabilities and Kaplan Meier curves
 - Incidence proportions, odds, incidence rates
 - Median event time
 - Age, period, cohort effects

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Open and closed cohorts

- A cohort is a defined group of individuals followed over time.
 - Remember, when we say “followed” this can happen in both a retrospective and prospective manner
- Closed cohort: has a fixed membership.
 - No one is added after cohort is formed
 - Cohort size remains the same (no LTFU) or decreases over time (due to LTFU)
 - Except if noted, we will generally be thinking about closed cohorts in this course
- Open cohort: can take on new members over time.
 - Aka a “dynamic” cohort
 - Can increase cohort size when enrollment is opened

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Prevalence

- A cross-sectional measure of disease occurrence
 - Prevalence count
 - Prevalence proportion
 - Prevalence odds
- Prevalence = proportion of a population that currently has the outcome over some period of time
 - What constitutes a ‘point of time’?
 - What constitutes a ‘period of time’?
 - Differences between what unit of time used often depends on health outcome being studied
 - BUT: Must ALWAYS provide a “period of time” over which prevalence is being measured!

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Prevalence counts

- Health insurance in the US (<https://www.census.gov/>):

2022	2023	2024
~304 million people	~305 million people	~310 million people

- Pros:**
 - Easy to understand
 - Communicates public health importance, assuming a well-understood condition and population denominator
 - Good for making an impact with impressive big numbers!
- Cons:**
 - Can't compare two counts unless denominator is same across populations/time periods (is denominator same from 2022 to 2024?)
 - Can easily be misused to create alarm: thousands get disease X a year! (But what's the denominator?)

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Prevalence proportions

- 1% of people in the US have type 1 diabetes ☒

OR

- 1% of people in the US developed type 1 diabetes ☐

- Pros:**
 - Communicates public health importance
 - Exposure prevalence is widely used in epidemiology
 - Similar* to incidence if duration of illness is very short
- Cons:**
 - Combines incidence and duration of conditions in question

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Odds, and of course prevalence odds

- For a given proportion P :

- odds(P)** = $P/(1-P)$

Proportion	Odds
0.02	$0.02/(1-0.02) = 0.0204$
0.05	$0.05 / (1-0.05) = 0.053$
0.10	$0.10 / (1-0.10) = 0.11$
0.25	$0.25 / (1-0.25) = 0.33$
0.33	$0.33 / (1-0.33) = 0.50$
0.50	$0.50 / (1-0.50) = 1.00$

- Prevalence odds** = $\text{Prev} / 1 - \text{Prev}$
- When the prevalence is nonzero, the prevalence odds will be numerically higher than the prevalence proportion.
 - When prevalence proportion is low ($<10\%$), prevalence odds is numerically similar to the prevalence proportion
 - When prevalence is high ($>10\%$), then prevalence odds overestimates prevalence proportion.

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Why does understanding this relationship matter?

- In 2024, conducted a study of diabetes among $n=2000$ adults in West Virginia
 - Identify 368 people with diabetes among the 2000 sampled in 2024
 - Prevalence proportion:** $368/2000 = 18.4\%$
 - We estimate that 18.4% of adult residents of WV have diabetes in 2024
 - Prevalence odds:** $(368/2000) / (1632/2000) = 22\%$
 - We estimate the odds of having diabetes compared to not having diabetes are 0.22, among adult residents of West Virginia in 2024. 🤔 😊

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Prevalence odds: pros and cons

- In general, for odds:
 - **Pros:**
 - Ratios of odds (odds ratios) have some useful properties (more later)
 - **Cons:**
 - Not intuitive 
 - Odds ratios can overstate effects if outcome is common
- Bottom line: if you see an odds or odds ratio presented, best to try and figure out the underlying proportion/percentages

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Survival data

- **Survival probability $[S(t)]$**
 - the probability that an individual survives from origin time (e.g. beginning of observation, diagnosis of cancer, etc.) to a specified future time t .
 - survival probabilities for different values of t provide summary information from time to event data
- **Survival curves:**
 - Theoretically, survival curves should be smooth functions
 - When using real data, the estimated survival curve is updated when only an outcome (e.g. death) occurs
 - Thus, in practice, they look more like step functions than smooth lines

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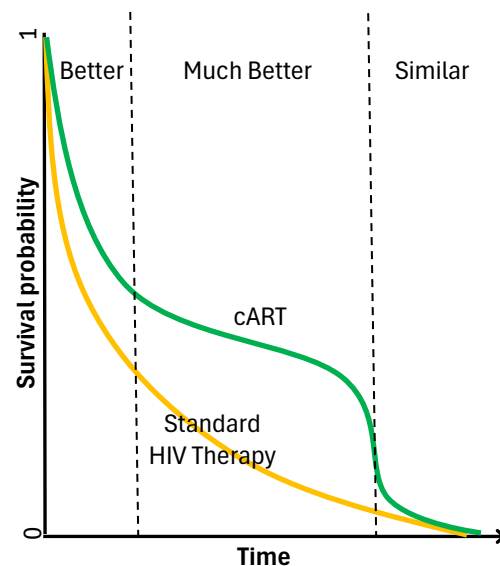
Survival probabilities

- What is $S(t)$ at the start of follow-up?
 - $S(0)=1$
- What is $S(t)$ at the end of follow-up?
 - Depends, but $S(LIFETIME)=0$
- How does $S(t)$ change with time?
 - Decreases or constant (never increases)
- Survival is $1 - \text{cumulative risk}$
- And $Risk_t = 1 - S(t)$

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Using survival probabilities to create survival curves

- Survival = $1 - \text{Risk}$, plotted over time
- Survival curves begin at 100% and move downwards over time
- Survival analysis is a powerful tool for understanding “time-dependent” effects
 - understanding how the probability of survival changes over time.
- Incidence risk can be considered a “slice” of a survival curve



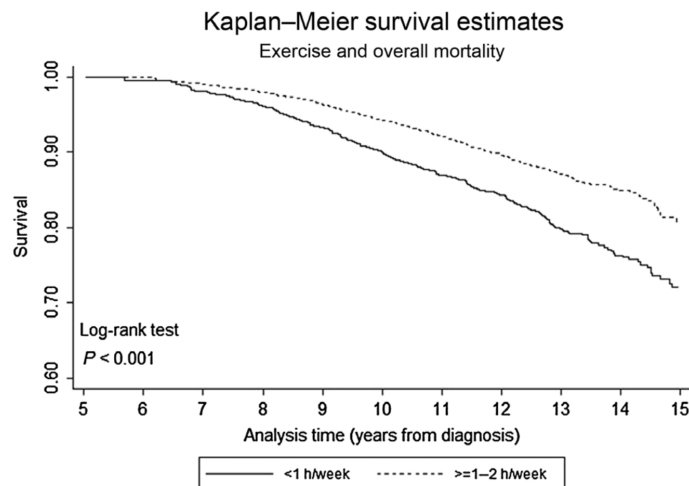
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Kaplan-Meier method for survival estimation

- Survival probability only changes at point when someone experiences event of interest (e.g. death, dx diagnosis, etc.)
- Can be used to determine important information about a single population such as median survival time
- To statistically compare multiple populations each with their own corresponding survival curve.
- Based on 3 assumptions:
 - patients who are censored have same survival probability as those who continue to be followed.
 - survival probabilities are the same for subjects recruited early and late in the study
 - that event happens at the time specified

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KM curve examples



Freedland SJ, Humphreys EB, Mangold LA, et al. Risk of Prostate Cancer–Specific Mortality Following Biochemical Recurrence After Radical Prostatectomy. JAMA. 2005;294(4):433–439. doi:10.1001/jama.294.4.433

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Standard incidence measures

- **Count:** # of new events in a given period of time
- **Risk:** probability of new events in a given period of time
 - AKA: incidence proportion or cumulative incidence
- **Odds:** $R/1-R$
- **Rate:** a count of events for a given amount of time experienced by a given number of people
 - AKA: incidence density

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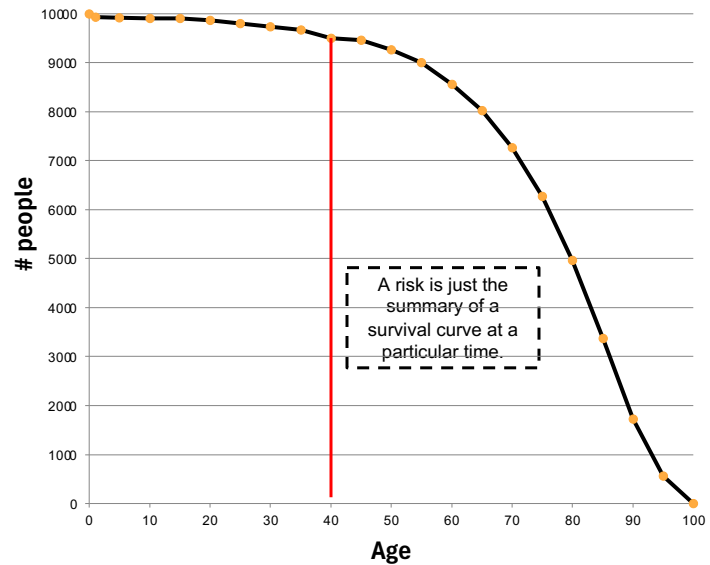
Reminder re: Risk

- Is a probability
 - what do we know about probability? What is its range?
 - Why the term 'incidence proportion' is preferred!
- Must have a fixed period of time
 - "Risk of ovarian cancer among women in the US is 1.1%"
 - can't mean a 1.1% risk in next 24 hours; a 1% risk would be high for 1 year (unless the women in question have other predisposing risk factors); over their lifetimes makes the most sense. (<https://seer.cancer.gov/statfacts/html/ovary.html>)

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Risk is a function of events over time

- 10,000 Americans followed from birth to age 100 (or death).
- What is the 40-year risk of death for these 10,000 Americans?
 - N at risk =
 - S surviving up to year 40 =
 - A = # of events ($N-S$) =
 - 40 year risk = A/N =



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Problems with estimating risk

- Poorly defined population
 - Lack ability to observe entire population
 - Lack ability to identify 'at risk' population
- Poorly defined observation period
 - People enter study at different times
 - People followed-up for different lengths of time by the end of study
- Incomplete follow-up over observation period
 - People are lost from your study.
- Sometimes we unable to distinguish incident and prevalent cases

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Communicating risk

- We need to do it, but we need to do it carefully and correctly!

HEALTH | 2/06/2014 @ 4:12PM | 3,685 views

Eating Yogurt Could Cut Diabetes Risk by 28 Percent, New Research Shows

+ Comment Now + Follow Comments

- What do you think that headline means? Explain!
- Communicating correctly and coherently BOTH matter!

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Dietary dairy product intake and incident type 2 diabetes: a prospective study using dietary data from a 7-day food diary

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Robert N. Luben · Kay-Tee Khaw ·
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Abstract

Aim/hypothesis The aim of this study was to investigate the association between total and types of dairy product intake and risk of developing incident type 2 diabetes, using a food diary.

Methods A nested case-cohort within the EPIC-Norfolk Study was examined, including a random subcohort ($n=4,000$) and cases of incident diabetes ($n=892$, including 143 cases in the subcohort) followed-up for 11 years. Diet was assessed using a prospective 7-day food diary. Total dairy intake (g/day) was estimated and categorised into high-fat ($\geq 3.9\%$) and low-fat ($< 3.9\%$ fat) dairy, and by subtype into yoghurt, cheese and milk. Combined fermented dairy product intake (yoghurt, cheese, sour cream) was estimated and categorised into high- and low-fat. Prentice-weighted Cox regression HRs were calculated.

fermented dairy product intake (T3 vs T1, HR 0.76 [95% CI 0.60, 0.99]; $p_{\text{trend}}=0.049$) and specifically with yoghurt intake (HR 0.72 [95% CI 0.55, 0.95]; $p_{\text{trend}}=0.017$) in multivariable adjusted analyses.

Conclusions/interpretation Greater low-fat fermented dairy product intake, largely driven by yoghurt intake, was associated with a decreased risk of type 2 diabetes development in prospective analyses. These findings suggest that the consumption of specific dairy types may be beneficial for the prevention of diabetes, highlighting the importance of food group subtypes for public health messages.

Keywords Cheese · Dairy · Fermented dairy · Milk · Type 2 diabetes · Yoghurt

Abbreviations

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Incidence proportion/risk: pros and cons

- Pros
 - It's an intuitive and clear measure – a probability makes sense!
- Cons
 - Interpretation becomes conditional for recurrent (non-terminal) outcomes
 - Does not account for variable length of follow-up between subjects. (BUT an incidence rate addresses this issue!)
 - Loses information compared to what a survival curve can provide!

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Odds: Also a function of risk

- For a risk R , odds are: $R / (1-R)$
 - If risk = 0.05, odds are
 $0.05 / (1 - 0.05) = 0.05 / 0.95 = 0.052$
- If a risk is 0.10, then odds = ...?
- If a risk is 0.20, then odds = ...?
- If a risk is 0.40, then odds = ...?
- Like risk:
 - Need defined population
 - Need fixed time period, and have to report it
 - Dimensionless (no units)
- Not like risk:
 - Range is 0 (when risk = 0) to ∞ (when risk = 1)
- Rule of thumb for odds
 - If the risk is $\leq 10\%$, then odds $\approx$ risk.
 - Otherwise, be very cautious: odds will tend to be more extreme than risk; and will tend to be misinterpreted as risk.

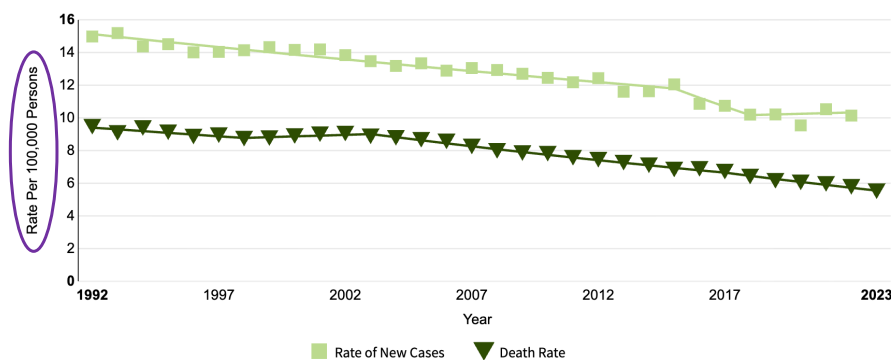
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Incidence rate

- Incidence rate (Incidence density) = frequency with which incident events occur within a given amount of follow-up time.
 - A/PT
 - Dimension is time^{-1} (1/time or inverse time)
 - Min = 0, Max = ∞
 - Inverse of incidence rate = average time to the event of interest
 - If: the number of people exiting the population due to death or emigration is identical to the number of people entering the population due to birth or immigration. In practice, this is unusual.

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Age-adjusted rates of new cases of Ovarian cancer and deaths per 100,000 person-years



New cases come from SEER 12. Deaths come from U.S. Mortality.

All Races, Females. Rates are Age-Adjusted.

Modeled trend lines were calculated from the underlying rates using the [Joinpoint Trend Analysis Software](#).

The 2020 incidence rate is displayed but not used in the fit of the trend line(s). [Impact of COVID on SEER Cancer Incidence 2020 data](#)

New cases are also referred to as incident cases in other publications. Rates of new cases are also referred to as incidence rates.

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Person-time

- Remember: person-time is the sum of time (t) that each individual i was at risk for developing the outcome: $\sum_i t_i$
- Some issues to keep in mind when considering ‘time at risk’, for example:
 - For risk of uterine cancer, person-time observed after a hysterectomy is not “at risk”
 - For infectious disease with acquired but partial immunity, person-time at risk may vary over time
- Sometimes a denominator which is not time, may be more appropriate
 - In motor vehicle accident studies, vehicle-miles driven is of interest
 - In cigarette use studies, pack-years smoked is of interest

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Population, fatal motor vehicle crashes, motor vehicle crash deaths and motor vehicle crash death rates per state, 2023

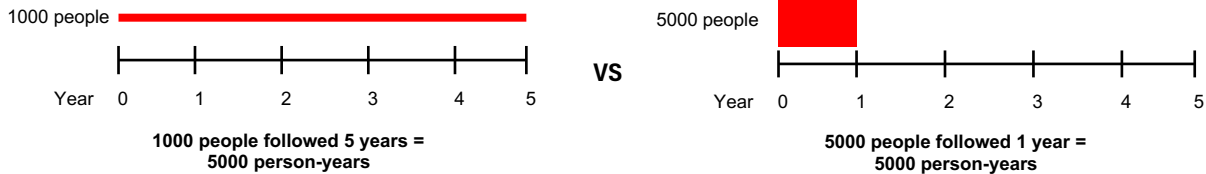
State	Population	Vehicle miles traveled (millions)	Deaths	Deaths per 100,000 population	Deaths per 100 million vehicle miles traveled
Alabama	5,108,468	72,289	974	19.1	1.35
Alaska	733,406	5,617	60	8.2	1.07
Arizona	7,431,344	75,297	1,304	17.5	1.73
New York	19,571,216	120,067	1,114	5.7	0.93
North Carolina	10,835,491	122,258	1,561	14.4	1.28
North Dakota	783,926	9,896	106	13.5	1.07
Ohio	11,785,935	113,224	1,242	10.5	1.10
Oklahoma	4,053,824	45,760	718	17.7	1.57
Oregon	4,233,358	36,816	587	13.9	1.59
Pennsylvania	12,961,683	100,126	1,211	9.3	1.21
Rhode Island	1,095,962	7,564	71	6.5	0.94
South Carolina	5,373,555	60,794	1,047	19.5	1.72

<https://www.iihs.org/>

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Person-time

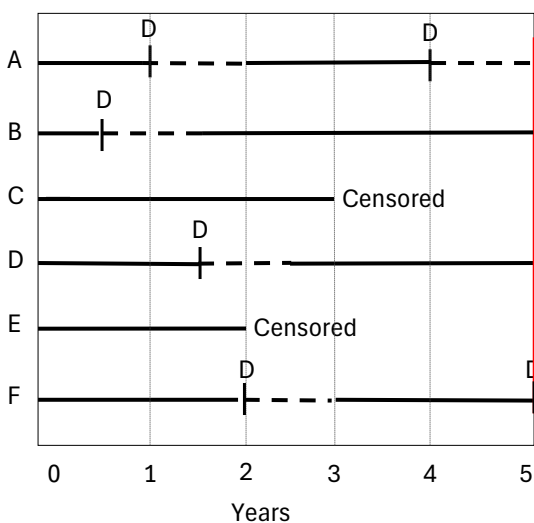
- First, person-time estimates can be misleading...



- Second, take pack-years of smoking = 1 pack a day for a year = 1 pack-year
 - 1 pack a day for 20 years = 20 pack-years
 - 4 packs a day for 5 years = 20 pack-years
- Do these two estimates have the same risk for an outcomes associated with them? Why? Why not?

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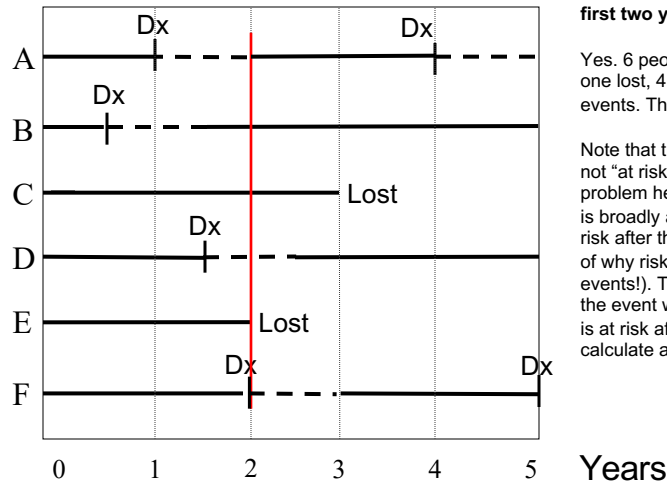
Handling person-time with recurring events



- People can have more than one diagnosis of D!
- After first diagnosis, immunity lasts for one year; then at-risk again
- Person-time:
 - A: 3 person-years at risk
 - B: 4 person-years at risk
 - C: 3 person-years at risk then lost
 - D: 4 person-years at risk (1.5 pre-diagnosis, 2.5 post)
 - E: 2 person-years at risk, lost
 - F: 4 person-years at risk (2 pre-diagnosis, 2 post)
 - Total: $3+4+3+4+2+4 = 20$ person-years at risk.
- Total events:
 - $2+1+0+1+0+2 = 6$ events
- Incidence rate:
 - $6/20$ person-years = 30 per 100 person-years

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Example (more than one event)



Can we calculate a risk over the first two years?

Yes. 6 people at risk at baseline, no one lost, 4 events, no repeated events. The two-year risk is 4/6.

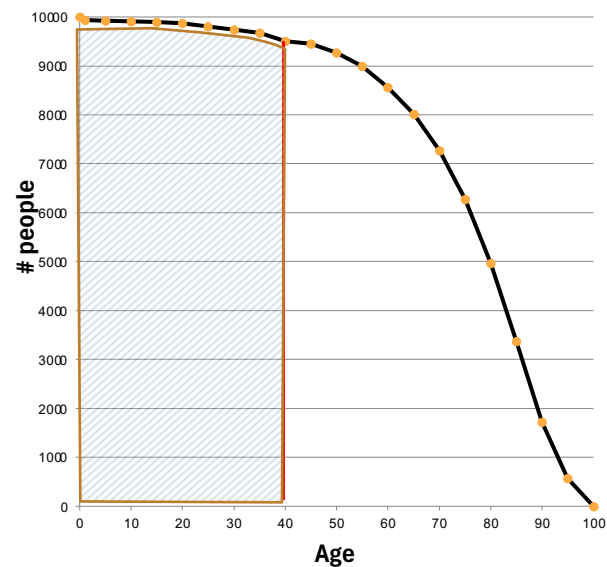
Note that the fact that people are not "at risk" after their event is not a problem here: in a risk calculation, it is broadly assumed that no one is at risk after the first event (this is part of why risks cannot handle recurrent events!). This is no different than if the event were death (where no one is at risk after it occurs); we can still calculate a risk of death.

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Incidence rate is a function of events over time

- What is the 40-year death **rate** for these 10,000 Americans?
 - There were 500 events.
 - How much person-time?
- Person-time is just people over time.*
- If no deaths or LTFU, total person-time=
 - $10,000 \times 40 = 400,000$ person-years
- Instead, it's 392,689 (based on people dying every year!)
 - Death Rate = $500 / 392,689 = 1.27$ deaths per 1000 person-years
 - 0.05 deaths/40 person-years
 - Similar to 40-year risk of 5%!



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Moving between incidence risk and incidence rate

- If risk > 10% (for example, when you want to estimate lifetime risk), can use the following formula:

$$R_t = 1 - e^{-\sum_i IR_i \Delta t_i}$$

- Where i indexes the intervals (e.g., 0-1 years, 1-5 years...), R is risk, IR is incidence rate, and Δt is length of the interval.
- Assumes:
 - Population is closed
 - The event under study is inevitable (no competing risks)
 - The number of events within each interval is a small proportion of the number at risk N at that time. That is, the risk at any time is small.
- Alas, there is a much simpler formula which holds in even more restrictive circumstances: risk $\approx$ incidence rate $\times$ time.

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Example

- What is the five-year risk of lung cancer when the rate is 8 cases/10,000 person-years?
 - $8/10,000$ (cases/p.y.) $\times 5$ (years) = 0.00400 = 0.40% in five years
 - The exponential formula yields 0.00399% –similar, because low risk
- Key message:
 - “Risk = incidence rate $\times$ time” is approximately valid only if Risk (IP) is <10%

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Incidence rate: pros and cons

- **Pros**

- Uses virtually all information available
- Readily compare populations or studies with different follow-up periods

- **Cons**

- Not as intuitive as other measures (to some)
- Strong assumptions about each unit of person-time being interchangeable

- What are some conditions under which we'd prefer an incidence rate rather than a risk?

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Incidence proportion vs incidence rate

- Risk cannot > 1 , incidence rate can > 1
- Risk has no units (i.e., it's a percent); incidence rate has units of inverse person-time
- Risk needs a fixed time period (ideally, over which all subjects were observed fully); incidence rate does not require a fixed time period of observation
- Risk has trouble handling recurrent outcomes; incidence rate does not have trouble handling recurrent outcomes

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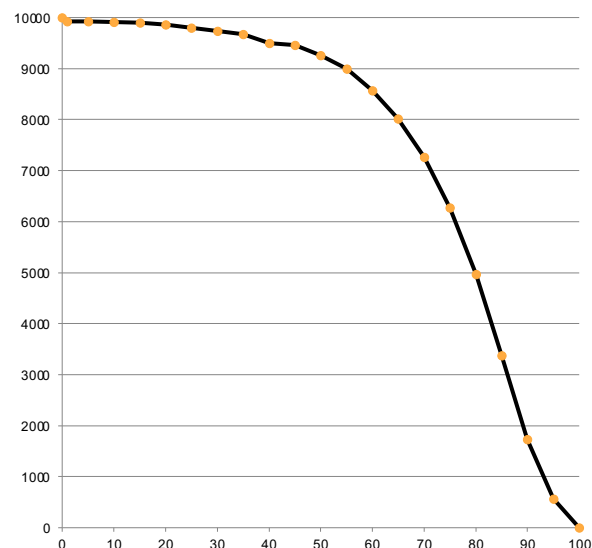
Factors to consider in measuring Incidence

- Competing events: events that preclude the event of interest from occurring and are theoretical if not practical problems for all outcomes other than all-cause mortality
- Left Truncation: aka late entries or people who are enrolled at a later point in your study
- Right-censoring: aka Loss to follow-up
- Can we assume there is some independence in left and/or right censoring?

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Median event time

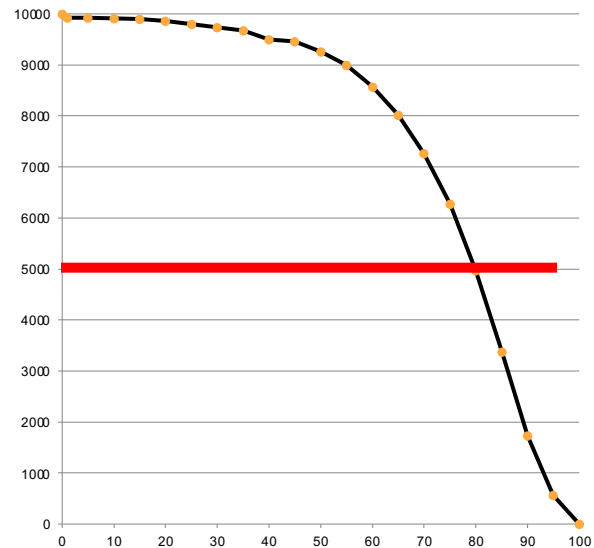
- Median is distinct from mean: how?
- All of our measures thus far have been
 - Based in the survival curve
 - Fixed in time: the line drawn across the survival curve is vertical
- Consider the following: at what time have half the people experienced the outcome?
 - That is, at what age are half the people dead?



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Median event time

- At what time have half the people experienced the outcome?
 - *At what age are half the people dead?*
- Estimate:
 - Draw line at 50% (in this case, at 5000 people)
 - At what age does the 50% line intersect the curve? ~80 years old
- Can estimate for any percentile!



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Median event time

- Often more informative than other measures
- Especially for inevitable outcomes like death
 - The risk for death on a long enough time scale is 1.
 - The median time to event (or some other percentile time to event) is sometimes more interesting.
 - If not even half the population gets the event, can report (for example) 25th percentile time to event instead of median (that is, 50th percentile).
- Underutilized as a measure, but good potential for communication of results to the general public – an important consideration!

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Defining and measuring exposures and outcome

- Want clear and precise, unambiguous definition of the outcome being described
- For survey question: timeframe(s) matter!!
 - Asking about cumulative lifetime?
 - “Ever” questions
 - What is the time period for a “period prevalence”?
 - 1 month (last 30 days)?, last 6-months? 1 year?
 - What is the time period for a “point prevalence”?
 - Today? This week? This month?
- For biological tests: is there a gold standard? If not, what is the Sn and Sp of the available test?

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What type of measure is most useful?

- To decide between incidence vs. prevalence, consider:
 - What are you able to measure?
 - Incidence (risk or rate) requires follow-up
 - Will we be using this info for building causal or predictive studies?
 - Prevalence only gives us info on burden
 - May be more relevant and useful and meaningful if our purpose is public health programing and policy

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Age, Period & Cohort Effects

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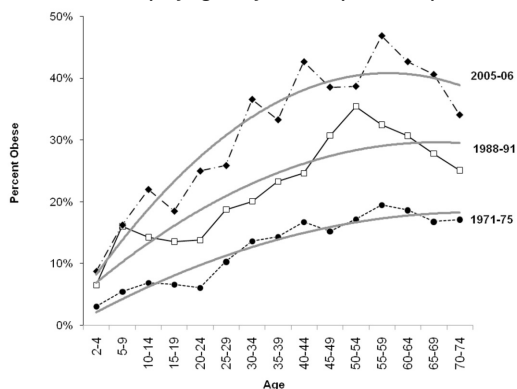
Age, Period, Cohort effects

- **Age effects:** describe variations in the outcomes associated with biological, social, or developmental processes of aging regardless of the time period or birth cohort to which an individual belongs
- **Period effects:** arise as a result of external factors (e.g. environmental, social and economic factors such as war, famine, economic crisis, etc.) that equally affect all age groups at a particular calendar time
- **Cohort effects:** variations resulting from the unique experience/exposure of a group of subjects (cohort) as they move across time.
 - most commonly defined group in epidemiology is the birth cohort based on year of birth
 - difference in the risk of a health outcome based on birth year

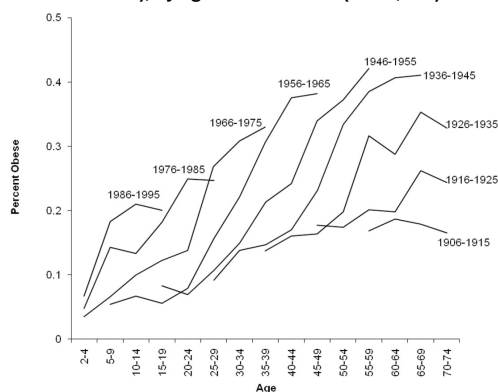
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Age, Period, Cohort effects (cont.)

Prevalence of Obesity in the United States (1971–2006), by Age & by Period* (N=91,755)



Prevalence of Obesity in the United States (1971–2006), by Age & Birth Cohort (N=91,755)



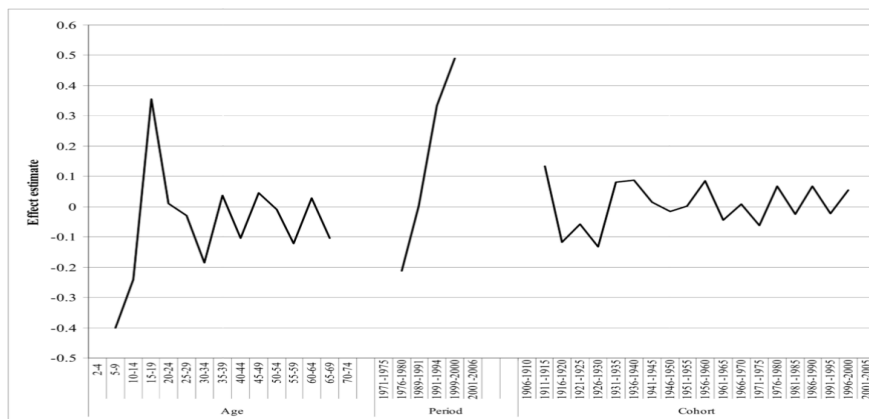
Keyes KM, Utz RL, Robinson W, Li G. What is a cohort effect? Comparison of three statistical methods for modeling cohort effects in obesity prevalence in the United States, 1971–2006. Soc Sci Med. 2010 Apr;70(7):1100–8.

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APC analysis

- APC analysis aims at describing and estimating the independent effect of age, period and cohort on the health outcome under study

Estimated curvature+ of age-period-cohort effects on obesity prevalence in the United States, 1971–2006 (N=91,755)



Keyes KM et al. Soc Sci Med. 2010 Apr;70(7):1100–8.

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